

Script for LinTeX User Study

Name of department: Computer and Information Sciences

Title of the study: LinTeX: Promoting Accessibility in LaTeX Documents

Introduction

Hello, I am Mohammad Danyal, a final year Computer Science student at the University of Strathclyde. I am the Researcher for this user study.

As a general overview of what this user study will consist of:

- Introductory questions to gauge experience levels
- An initial phase for listening to the document through a screen reader.
- Answering questions about how easy or difficult the document is to understand.
- Utilising the proposed tool, LinTeX, to make the document more accessible to screen reader users.
- A final listening of the document, with the changes requested by the tool having been applied.
- Answering further questions about whether the tool had made an impact on the difficulty for understanding the document as a screen reader user.

These tasks will be explained in more detail as we approach them.

If you have any questions as we go through the process, feel free to chime in.

You have the right to opt to not answer questions as you wish, as stated in the consent form.

Introductory questions

- On a scale of 1 to 5, how often do you use LaTeX?
 1. Never
 2. Rarely
 3. Sometimes
 4. Often
 5. Almost always
- On a scale of 1 to 5, how often do you use screen readers?
 1. Never
 2. Rarely
 3. Sometimes
 4. Often
 5. Almost always
- On a scale of 1 to 5, how familiar are you with listening to audio recordings of speaking?
 1. Very unfamiliar
 2. Unfamiliar
 3. Neither unfamiliar/familiar
 4. Familiar
 5. Very familiar

Initial listening

For this step, I will have you listen to the output of the document before any changes have been made to it. If you have any device for listening to the output that you'd prefer to use, just make that preference known and it can be setup.

While listening to the output, if you would like to make any points of interest, simply indicate that and the audio output can be paused. Additionally, if you would like to repeat any sections or the full recording, feel free to ask.

When the recording has been complete, and no requests are made to repeat a section, we will move onto answering a few questions.

Questions [initial listening]

- Provide a general overview of what the document discussed.
- What parts of the listening were simple to understand? Why?
- What parts of the listening were difficult to understand? Why?
- On a scale of 1 to 5, how would you rate the quality of the audio output? (i.e. in terms of the content of its output)
 1. Very poor
 2. Poor
 3. Neither poor/good
 4. Good
 5. Very good
- How well could you understand the purpose of tables, images or figures in the listening?

Using LinTeX

In this step, we will perform a think aloud session. In case you have not participated in one before, here is a general summary:

- You will be tasked to perform some action.
- You should audibly state your thought process as you go through the task. For example, this may be in the form of the stating having trouble with something.
- There will be minimal intervention from me.

The task is in 3 parts:

1. Upload the document to LinTeX by navigating its GUI
2. Improve the document's accessibility by following what LinTeX provides.
3. Export the document.

You will be provided with the file's location, as well as the LinTeX tool to work with.

Final listening

Similar to before, you will be tasked with listening to the output of the document. This time, after changes have been applied through the LinTeX tool.

If you would like to make any points of interest, simply indicate that and the audio output can be paused. Additionally, if you would like to repeat any sections or the full recording, feel free to ask. When the recording has been complete, and no requests are made to repeat a section, we will move on to answering the final questions.

Questions [final listening]

- What parts of the listening were simple to understand? Why?
- What parts of the listening were difficult to understand? Why?

- On a scale of 1 to 5, how would you rate the quality of the audio output? (i.e. in terms of the content of its output)

1. Very poor
2. Poor
3. Neither poor/good
4. Good
5. Very good

- How well could you understand the purpose of tables, images or figures in the listening?

- On a scale of 1 to 5, how likely are you accommodate for screen reader users in the future?

1. Very unlikely
2. Unlikely
3. Neither unlikely/likely
4. Likely
5. Very likely

Exit

Thank you for your time. You have been of great help for this user study. If you wish to see where the project may end up or have any questions, my email is on the Participant Information Sheet under the Researcher.